Designing a Gamified Application for User Privacy Preference Extraction

Continual Assessment Report

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Contents

1 Introduction **2**
- 1.1 Background . . . 2
- 1.2 Project objectives . . . 2

2 Ideation of the Solution **3**
- 2.1 Review of current cookie consent practices . . . 3
  - 2.1.1 Exhibit A: F1 official website . . . 3
  - 2.1.2 Exhibit B: FIFA official website . . . 4
- 2.2 Requirements . . . 5
- 2.3 Constructing the solution . . . 5
  - 2.3.1 Concept . . . 5
  - 2.3.2 The “Home” analogy . . . 6
  - 2.3.3 Chatbot interface . . . 6
  - 2.3.4 Dynamic questions . . . 6
  - 2.3.5 Translating user answers into decisions . . . 6
  - 2.3.6 Accessibility and appearance . . . 6
  - 2.3.7 Addressing the requirements . . . 6
- 2.4 Constraints . . . 7
  - 2.4.1 Requesting permissions . . . 7
  - 2.4.2 Limited CMP support . . . 7
  - 2.4.3 Handling data safely . . . 7

3 Literature Review **8**
- 3.1 CookieBlock: Automating Cookie Consent and GDPR Violation Detection . . . 8
- 3.2 Research gap and implications for this project . . . 8

4 Progress to Date **9**
- 4.1 Work completed . . . 9
  - 4.1.1 Browser extension adapter . . . 9
  - 4.1.2 “Home” analogy interface . . . 10
- 4.2 Design alternatives . . . 10
- 4.3 Challenges and risks . . . 11
  - 4.3.1 Limited CMP support . . . 11
  - 4.3.2 Uncertain cookie purposes . . . 11
  - 4.3.3 Natural-language misinterpretation . . . 11

5 Research Plan for Next Semester **12**
- 5.1 Planned work . . . 12
- 5.2 Milestones . . . 14
- 5.3 Evaluation and success criteria . . . 14

Acknowledgement of AI Use **18**

Chapter 1

Introduction

1.1 Background

Despite rapid advances in technology, privacy preferences remain an unfamiliar topic for many members of the general public. Online privacy preferences can be collected in several ways, including through surveys [1], the parsing of device settings [2], and interactions with cookie consent banners [3]. This project focuses on eliciting user privacy preferences in relation to cookies, whose use is commonly disclosed through pop-up banners provided by Consent Management Platforms (CMPs) when users visit a website [4]. These banners ask users to indicate their consent preferences for cookies. Cookies are small blocks of data stored on a user's device that allow websites to retain state information about the user or their browsing activity [5].

Most users pay little attention to the privacy choices they are asked to make when visiting a new website and encountering a cookie banner. When presented with a pop-up containing lengthy text, they may be inclined to dismiss it quickly [6]. Combined with limited knowledge of privacy preferences and how to manage them, this may lead users to choose whichever option lets them proceed immediately, often the "Accept All" or "Deny All." options The complexity of cookie banners and users' knowledge limitations may help explain why users overlook the choices they present [7].

In response to this problem and its possible causes, this project explores whether gamification can make the process of extracting privacy preferences through cookie banners more approachable.

1.2 Project objectives

Based on this background, the project has three main objectives:

1. Investigate existing solutions that address similar issues.
2. Design a solution that presents cookie management through a gamified interface.
3. Evaluate whether a gamified process can help users manage their privacy preferences on their own initiative.

Chapter 2

Ideation of the Solution

Before developing the proposed solution, it is useful to review common practices for collecting users' cookie consent preferences from some websites.

2.1 Review of current cookie consent practices

As noted in the introduction, most websites use specialised tools called Consent Management Platforms (CMPs) to record users' consent choices for cookies [8]. These platforms commonly present a pop-up asking users to indicate their preferences when the browser has no stored record of users' cookie consent choices for that website.

2.1.1 Exhibit A: F1 official website

The official F1 website [9] provides an example of this process. When users first visit the website, they are greeted by the pop-up shown in Figure 2.1.

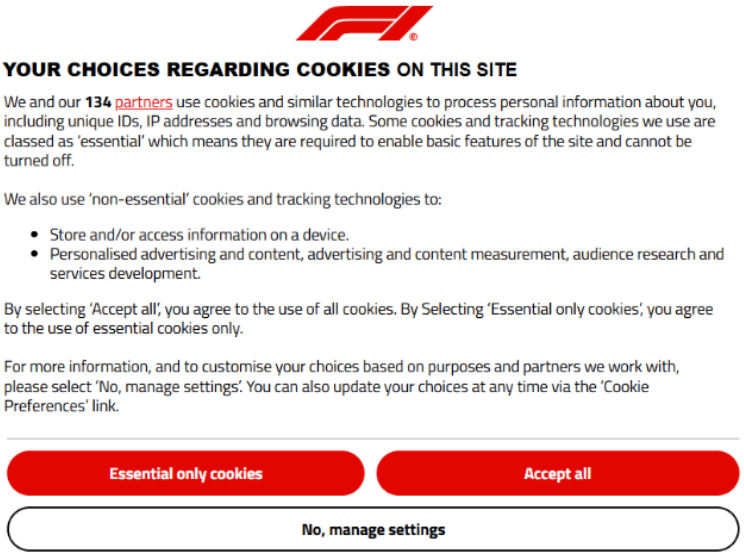


Figure 2.1: Cookie consent banner displayed on the F1 official website.

The interface presents three main choices: accept only essential cookies, accept all cookies, or open the settings to manage consent preferences. Although the interface provides ample information about cookies and their purposes, the wording may appear lengthy or difficult for a general audience to understand. The red highlighting may also draw users' attention towards

options that permit more cookies. This is an example of a dark pattern: a manipulative user experience design that can steer users towards choices that benefit the company behind [10].

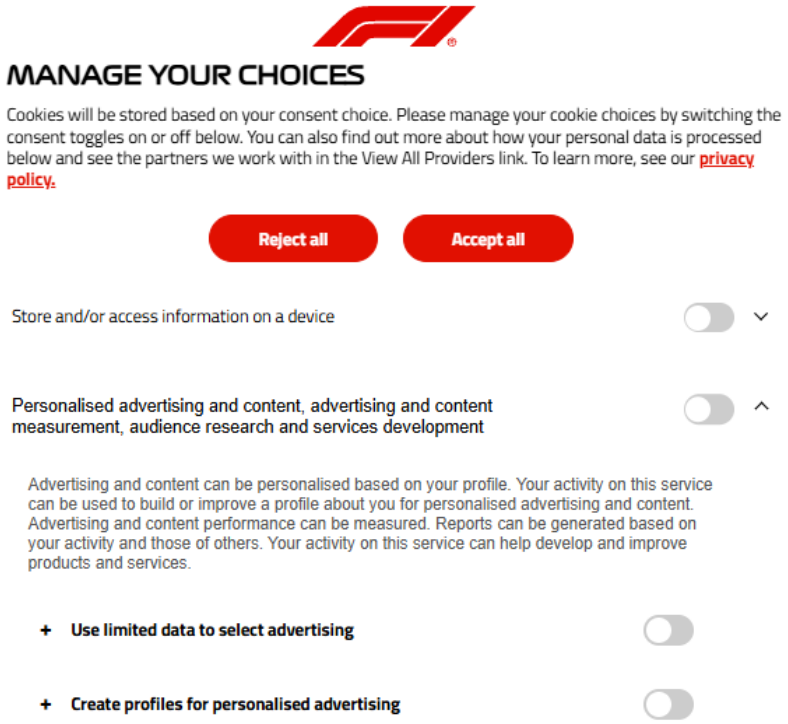


Figure 2.2: F1 cookie settings window for managing consent preferences.

After selecting "No, manage settings", users can indicate the purposes for which data collected through cookies may be used. These purposes include selecting advertisements and creating profiles for personalised advertising, as shown in Figure 2.2. Although this gives users more control, it also adds complexity because they must consider each option and decide whether to consent.

2.1.2 Exhibit B: FIFA official website

The official FIFA website [11] provides another example. Its cookie banner offers users three buttons for indicating their preferences, as shown in Figure 2.3.

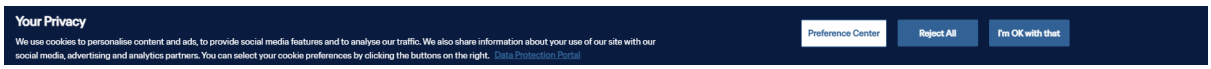


Figure 2.3: Cookie consent banner displayed on the FIFA official website.

If users select the white "Preference Center" option, they are taken to the interface shown in Figure 2.4. Cookies that require consent, such as functional cookies, are grouped together in one window. This may make it easier to customise preferences, but it offers less granular control: users cannot consent to selected functional cookies while rejecting others in the same category. By comparison, F1's settings interface in Figure 2.2 lets users indicate preferences for different cookie purposes. FIFA's interface is nevertheless more concise and easier to review.

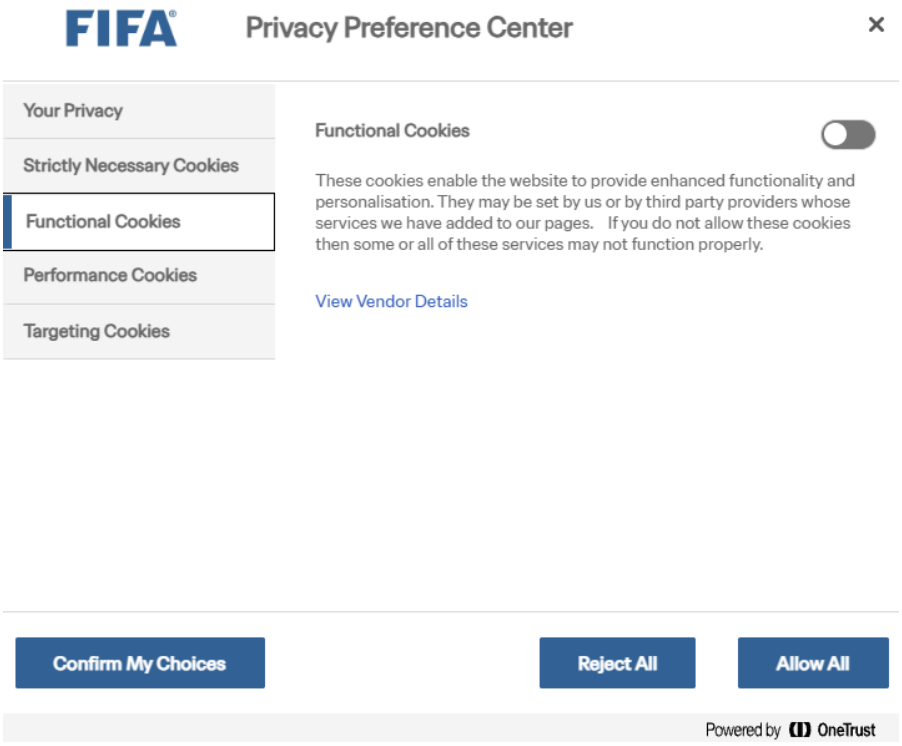


Figure 2.4: Privacy preference centre on the FIFA official website.

2.2 Requirements

The review of current cookie consent practices suggests five requirements for the proposed solution:

1. **Conciseness:** Users should not have to read lengthy content to understand the implications of their privacy choices; the interface should minimise confusion.
2. **Convenience:** Users should be able to access and set up the application easily.
3. **Engagement:** The process of eliciting privacy preferences should hold users' attention.
4. **Information:** The application should provide sufficient information about how users' privacy choices affect them.
5. **User autonomy:** Users should retain as much control as possible over their cookie consent preferences.

2.3 Constructing the solution

The proposed solution is a privacy companion that explains the choices a website requests and turns privacy decisions into an interactive, scenario-based experience.

2.3.1 Concept

Users encounter privacy choices whenever they visit websites, yet many overlook them. One reason may be that current consent interfaces can feel dull, lengthy, and inconvenient; encountering many such pop-ups may be tiring [7]. This project therefore explores a gamified application that simulates the process of managing privacy preferences. It aims to simplify technical details

and make them more approachable. Visual representations of cookies, third parties, and privacy choices may also help engage users and present the subject in a more attention-grabbing way.

2.3.2 The "Home" analogy

Browsing websites can be compared to visiting people's homes. Instead of presenting websites and cookies only through technical terms, the application can describe websites as homes that users visit and cookies as guests in those homes. Guests can then be personified according to each cookie's purpose. For example, an authentication cookie that lets a website recognise a returning user could be represented as the homeowner wanting to remember the visitor's name so they can enter again more easily [12]. This analogy can be presented through scenarios in which users indicate their cookie preferences.

If users want more information about a specific cookie, the application should also let them inspect its original details without relying on the analogy or scenario-based presentation. This provides access to information in a form closer to a conventional cookie consent interface.

2.3.3 Chatbot interface

Binary choices, like those often presented in cookie banners, may not capture the nuances of users' privacy preferences. A natural-language interface could give users more room to express those preferences. This project therefore uses a chatbot through which users can communicate with the application. The chatbot interprets their answers and translates them into decisions.

2.3.4 Dynamic questions

Websites use different cookies and request different forms of consent, so a fixed set of scenarios may not cover every cookie. The solution should generate scenarios dynamically from information about each cookie and its purpose. Dynamic scenarios can also serve as clarifying questions when a user's response is unclear and more information is needed before it can be translated into a deterministic preference decision.

2.3.5 Translating user answers into decisions

The backend processes users' chatbot responses to derive deterministic decisions. The application then applies those decisions and communicates users' preferences to the relevant CMPs.

2.3.6 Accessibility and appearance

The solution should be easy to set up and use. A browser extension would provide a convenient way to access the application. It should collect cookie data, support user interaction, and communicate with the CMPs used by websites. It should also provide a visual representation of users' current privacy preferences through the analogy described in Section 2.3.2.

2.3.7 Addressing the requirements

The proposed design addresses the requirements as follows:

1. **Conciseness:** The application simplifies lengthy information into scenario-based questions.
2. **Convenience:** A browser extension provides access in the same browser window and tab as the website.
3. **Engagement:** Visual representations and the chatbot use analogies to make privacy preferences and cookies more approachable and immersive.

4. **Information:** Users can choose to inspect further details when they want more information, while keeping those details optional.

5. **User autonomy:** Users can change their preferences by interacting with the chatbot.

2.4 Constraints

Several constraints and concerns are foreseeable at this stage of development.

2.4.1 Requesting permissions

Accessing cookie data requires the application to request permission from the user. The application should request only the permissions it needs and should handle permission requests carefully.

2.4.2 Limited CMP support

Websites use different CMPs and cookies, and CMPs have distinct APIs and cookie categories. The solution's reach will therefore depend on how many CMPs it supports. This project will focus on Cookiebot [13].

2.4.3 Handling data safely

Chatbot messages are sent to an external natural-language AI service. The application must handle this data carefully to reduce the risk of exposing users' private information. Users should be informed about this data flow and advised not to enter sensitive personal information.

Chapter 3

Literature Review

3.1 CookieBlock: Automating Cookie Consent and GDPR Violation Detection

One related solution is CookieBlock, which addresses cookie classification and consent management [14]. Like this project, CookieBlock takes cookie information as input and classifies cookies into categories. It uses a trained classifier and mappings between cookies and their purposes to determine whether cookies align with users' privacy preferences and should be accepted or rejected. The classifier predicts cookie purposes from attributes such as name, domain, and path. CookieBlock is also delivered as a browser extension, a format this project aims to adopt for convenience.

3.2 Research gap and implications for this project

This project uses browser cookie metadata, CMP signals, and cookie-purpose mappings to categorise cookies and incorporate them into scenario-based questions through the personification analogy. CookieBlock notes that cookie-purpose mappings in website privacy policies may be unreliable because policies vary between websites. In response, this project narrows its scope to Cookiebot declarations found in website HTML. It takes a more grounded approach by relying on these declarations rather than training a classifier to map cookies to their purposes, as CookieBlock does [15], [16].

Another reason a trained classifier may be unsuitable for this project is the risk of misclassifying a cookie's purpose, which CookieBlock also raises in its documentation. This project seeks to avoid that risk and maintain the reliability of the information it presents to users.

Concrete cookie-purpose mappings are needed to generate dynamic scenarios. The project does not aim to classify every cookie's purpose in real time. Instead, it is scoped to look up purposes in Cookiebot's scanned database and declarations, rather than to identify cookies across websites that use different CMPs. Its focus is to elicit user preferences through a natural-language interface that presents dynamic privacy scenarios using an accessible analogy. The project will explore whether this approach can make preference elicitation easier and increase users' interest in cookie consent management through gamified interaction, an issue that CookieBlock addresses only indirectly.

Chapter 4

Progress to Date

4.1 Work completed

The project has progressed towards representing privacy choices through the "Home" analogy described in Section 2.3.2. Two main features have been implemented: a browser cookie data adapter and the "Home" analogy interface. Together, they allow the application to generate dynamic, scenario-based questions from cookie data collected by the adapter.

4.1.1 Browser extension adapter

Generating dynamic scenarios based on cookie purposes requires a cookie input adapter. The adapter standardises cookie metadata and CMP signals, then categorises cookies by purpose. Without this step, the project cannot generate the text-based scenarios needed to elicit users' privacy preferences.

The browser extension adapter can collect cookie data from websites that use Cookiebot as their CMP. The collected data include:

1. **Cookie metadata:** Properties such as the cookie's name, lifespan, security attributes, and scope, which describe how it may be stored and transmitted and when it expires [17], [18].
2. **CMP signal data:** The user's existing consent choices for cookie categories on the website [19].
3. **Public cookie declaration:** A published list mapping cookies used on the website to their purposes [20].

The adapter uses the public declaration to map cookies to purposes, then presents them as "guests" in the "Home" analogy (Section 2.3.2).

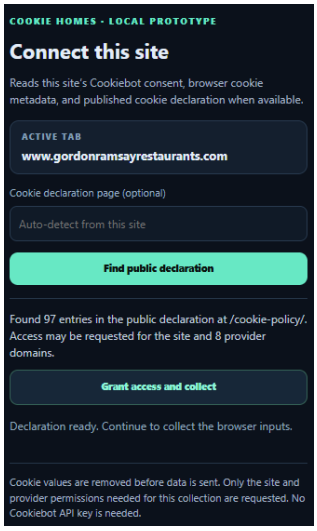


Figure 4.1: A browser extension cookie data adapter.

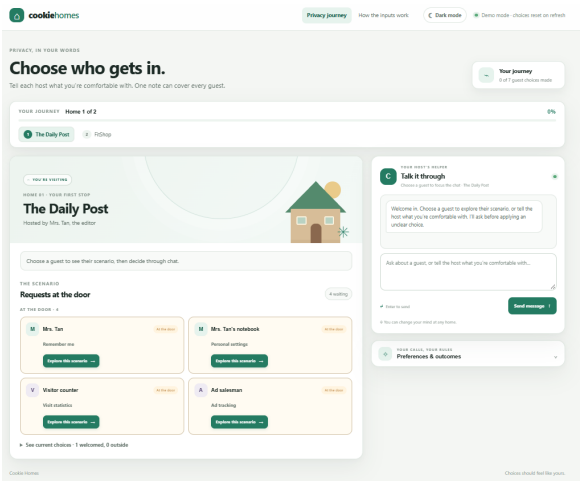


Figure 4.2: The main page of Cookie Homes.

4.1.2 "Home" analogy interface

A functional prototype, Cookie Homes, demonstrates the proposed scenario-based journey in which users express their preferences through questions and an AI chatbot. Users can discuss cookie purposes through a natural-language interface, receive clarification when a request is ambiguous, and apply choices through a deterministic rules engine. The prototype includes the home-and-guest analogy, progress tracking, and a summary of choices. It demonstrates the interaction concept but does not yet apply choices to a website's CMP. Cookies are personified as guests in each home's website scenario.

4.2 Design alternatives

Three interface approaches were considered and compared using clarity, user control, engagement, and implementation feasibility as criteria.

Table 4.1: Comparison of alternative interfaces for privacy preference elicitation.

Approach	Advantages	Limitations
Cookie category dashboard	Familiar controls give users direct choices for categories such as statistics and marketing. It is straightforward to implement.	Users still need to understand category labels, and the interface may feel similar to a conventional consent panel.
Fixed-choice scenario cards	Short descriptions make requests easier to scan, and allow/deny buttons keep decisions predictable without AI interpretation.	Choices are mostly binary and provide limited support for questions, explanations, or nuanced preferences.
Conversational home-and-guest interface (selected)	The home analogy makes cookie purposes more approachable. Conversation lets users ask questions, explain preferences, and clarify choices.	Natural-language interpretation can be incorrect, so clarification and rule-based validation are needed. The AI service also adds implementation and data-handling considerations.

The conversational home-and-guest interface was selected because it best supports the project's aims: making privacy choices understandable and engaging while allowing users to express preferences in their own words. The prototype addresses some of its risks by asking follow-up questions when intent is unclear and passing clear choices through a deterministic rules engine. The interaction is implemented, but user evaluation remains future work.

4.3 Challenges and risks

4.3.1 Limited CMP support

As discussed in Section 2.4.2, websites use different CMPs. The current prototype supports Cookiebot, which is sufficient to demonstrate the concept but limits the number of websites on which it can be used. Supporting other CMPs would require further work to collect consent signals and map cookies to their purposes.

4.3.2 Uncertain cookie purposes

The current cookie-purpose mapping relies on public cookie declarations, which may not be available for every website. To reduce the risk of misleading users, the prototype keeps unmatched cookies unclassified rather than inferring their purpose from cookie names alone. It can show the available information, such as cookie metadata, and let users make a choice about an unclassified cookie in the current scenario.

4.3.3 Natural-language misinterpretation

The chatbot uses an external natural-language AI model to interpret users' intent. Responses may be ambiguous or unclear, creating a risk that the model will misinterpret what a user means. The application should ask clarifying questions when needed and should not act until it has a sufficiently clear understanding of the user's decision.

Chapter 5

Research Plan for Next Semester

5.1 Planned work

The next stage will focus on completing a safe, usable end-to-end prototype and evaluating whether its gamified interface helps users understand cookie purposes, engage with consent choices, and make informed privacy decisions. The evaluation will examine which elements of the experience are helpful, whether they introduce any confusion or extra effort, and how the approach compares with more conventional ways of presenting cookie choices. Work will be prioritised so that a single-CMP extension and a controlled user evaluation are completed before attempting broad CMP coverage.

1. **Move the journey into a browser extension.** Rework the current website interface into an extension side panel or similarly persistent browser view, so users can review a site and make choices without switching to a separate local website. Keep site access permission-based and request only the access needed for the active site and supported CMP. Test installation, navigation between sites, permission denial, and recovery from connection errors. The expected output is an installable prototype with clear onboarding and privacy notices.

2. **Investigate and implement consent write-back.** Determine which documented Cookiebot browser-side mechanism, if any, can update a user's category choices on the active site. Implement and verify write-back for the supported categories, with a review step before applying choices, a clear success or failure message, and an undo or recovery path where technically possible. Essential categories must remain protected. If a supported, stable mechanism is unavailable, provide a transparent hand-off to the site's consent controls and document this as a limitation rather than modifying cookies directly. The expected output is a tested, narrowly scoped consent integration.

3. **Improve purpose evidence and uncertainty handling.** Continue to use public declarations as one source, but expose the evidence behind each mapping and distinguish confirmed, ambiguous, and unmatched records. Test duplicate cookie names, missing or stale declarations, absent Cookiebot snapshots, and mismatched domains. Keep uncertain records unclassified and let users inspect the available metadata before deciding.

4. **Extend the home-and-guest analogy.** Add optional explanations for relevant attributes such as who provides a cookie, whether it is first- or third-party, how long it lasts, and what its security attributes mean. Retain the original technical details as inspectable evidence, and avoid implying that an analogy proves a cookie's purpose or safety. Use brief formative feedback sessions to check whether the new explanations improve understanding without making the journey too long.

5. **Strengthen decision and data safeguards.** Add a concise review of the choices the system is about to apply and an undo path for supported changes. Test that ambiguous messages do not change state and that essential cookies cannot be rejected. Minimise the chat context sent to the external AI service, explain this processing to users, avoid collecting sensitive participant information, and document data retention and deletion.

6. **Evaluate the gamified interface with users.** After a small pilot, conduct a task-based study comparing the home-and-guest interface with a conventional category dashboard or fixed-choice scenario cards. Counterbalance the order of interfaces where practical. Before the tasks, record each participant's direct preferences using a short questionnaire and compare these with the choices expressed through each interface. Measure task completion, time, comprehension of cookie purposes and consequences, decision confidence, clarification frequency, and usability, trust, and engagement. Ask participants which elements of the analogy, scenarios, chatbot, and progress feedback helped or hindered them, and whether the gamified presentation encouraged them to pay more attention to cookie choices.

7. **Check accessibility and technical reliability.** Review keyboard-only use, screen-reader labels, contrast, readable language, and responsive layout. Add or refine automated checks for cookie normalisation, consent write-back safeguards, ambiguous AI replies, and essential-cookie invariants, then run end-to-end checks in the target browser. Record unsupported sites and failure cases.

5.2 Milestones

Table 5.1: Provisional next-semester milestones.

Period	Activity	Expected output
Weeks 1–2	Compare interface baselines, review CMP write-back options, identify the gamified elements to evaluate, and prepare the study protocol.	Agreed interface comparison and study protocol submitted for required review.
Weeks 3–5	Build the extension interface, implement the supported Cookiebot write-back path, and refine the analogy and scenario journey.	Installable prototype with reviewed choices, clear explanations, and documented limitations.
Weeks 6–7	Pilot the interface tasks and questionnaires; fix major usability, accessibility, and technical issues.	Pilot findings and revised evaluation materials.
Weeks 8–10	Run the user study and compare the gamified interface with the selected baseline.	Evaluation dataset and analysis of comprehension, decision quality, engagement, usability, and feedback.
Weeks 11–12	Refine the prototype, document limitations and future CMP support, and complete the report and demonstration.	Final prototype, demonstration, and report.

5.3 Evaluation and success criteria

Evaluation will address whether the prototype works as intended and whether its gamified presentation helps users understand and manage cookie choices. The study will compare the home-and-guest interface with at least one baseline, preferably a category dashboard or fixed-choice scenario interface, using the same privacy scenarios. Participants' stated preferences from a direct questionnaire will serve as a reference when assessing the choices captured by each interface. The evaluation will also examine whether users find the gamified experience more engaging and understandable, and whether it supports informed decisions without adding confusion or effort.

- **Preference extraction accuracy:** report how often the choices produced by each interface agree with participants' stated preferences. Separately record AI misinterpretations, cases correctly clarified, and cases where the system changed state despite ambiguity.
- **Understanding and decision quality:** compare comprehension of cookie purposes and consequences, task completion, time, decision confidence, and agreement between interface choices and questionnaire responses. Use participant comments to explain where the analogy or technical details helped or confused them.
- **Engagement and usability:** compare perceived interest, attention, clarity, effort, and willingness to use the interface between the gamified approach and the baseline. Ask participants which specific elements contributed to or detracted from their engagement.
- **Safety and reliability:** require that ambiguous input causes no preference change, essential categories cannot be rejected, uncertain purpose matches remain visibly uncertain, and consent write-back reports the actual outcome. Verify that cookie values are not collected or transmitted by the extension.
- **Accessibility and scope:** document results from keyboard and screen-reader checks, supported browser and CMP combinations, permission failures, and recovery from integration errors.

The project will be considered to have met its prototype objective if users can complete the supported consent journey, inspect the evidence behind a request, and correct or recover from an unintended choice. The effectiveness of the gamified interface will be assessed by comparing comprehension, decision quality, engagement, and usability with the baseline; results that show no improvement will still be reported as findings. Before the main study, the exact measures, analysis method, and any numerical targets will be agreed with the supervisor and recorded in the study protocol. A small exploratory sample can identify problems and provide preliminary comparisons, but will not support broad claims about all users.

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Acknowledgement of AI Use

I used OpenAI Codex (GPT-6) to help prepare the LaTeX report template and format the document, including arranging and sizing figures. I also used it to generate ideas, assist with drafting and refining wording in parts of the report, review grammar, explain project-related concepts, and identify relevant online sources. I reviewed and edited the generated suggestions and take responsibility for the accuracy, quality, and final content of this report. I also used it to help me write this declaration of AI use in my CA report.